

29/01/25

Lecture 8

Mean field Approximation of Ising Model

$$\mathcal{H} = -\frac{1}{2} \sum_{i,j} J_{ij} S_i S_j + h \sum_i S_i$$

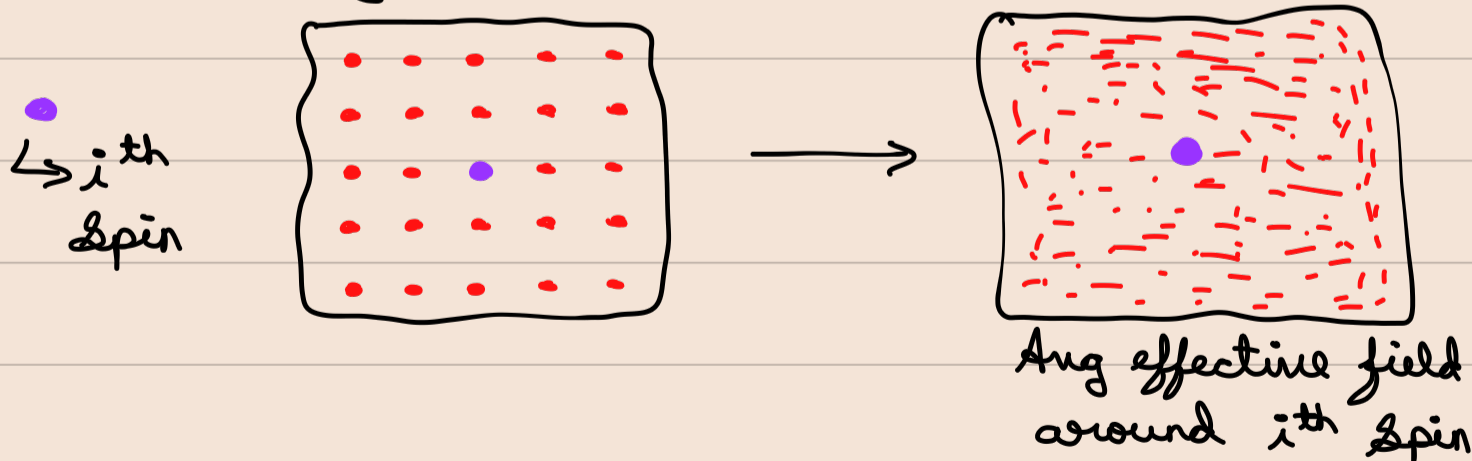
↳ (To remove double counting)

$$\Rightarrow \mathcal{H} = -\frac{1}{2} \sum_i \left(\sum_j \left[J_{ij} S_j S_i + h S_i \right] \right)$$

$$\mathcal{H} = -\frac{1}{2} \sum_i \left[\sum_j J_{ij} S_j + h \right] S_i$$

Mean field approx. is replacing the factor

" $\sum_j J_{ij} S_j + h$ " by its average



$$S_i S_j \xrightarrow{\text{M.F.T}} S_i \langle S_j \rangle + \langle S_i \rangle S_j - \underbrace{\langle S_i \rangle \langle S_j \rangle}_{\text{Can be ignored as it won't affect dynamics}}$$

Can be ignored as it won't affect dynamics

$\langle S_j \rangle = m$ (due to lattice translation symmetry the index disappears)

$$\Rightarrow \mathcal{H} = -\frac{1}{2} \sum_{ij} J_{ij} [S_i \langle S_j \rangle + \langle S_i \rangle S_j] - h \sum_i S_i$$

$$= -\frac{1}{2} \left[m \sum_{i,j} J_{ij} S_i + m \sum_{i,j} J_{ij} S_j \right] - h \sum_i S_i$$

Identical terms (gives factor of 2)

$J_{ij} = J_{ji} \Rightarrow$ Symmetry in Exchange energy

$$\Rightarrow \mathcal{H} = -\frac{1}{2} \left[2 m \sum_{ij} J_{ij} S_i \right] - h \sum_i S_i$$

$$\mathcal{H} = -m \sum_{ij} J_{ij} S_i - h S_i = -\sum_i S_i \left[\underbrace{\sum_j J_{ij} m}_{H_i} + h \right]$$

Let H_i = effective average field experienced by the spin at i^{th} lattice. Then,

$$H_i = m \sum_j J_{ij} + h$$

Defining $\sum_j J_{ij} = g_i \Rightarrow H_i = m g_i + h$

where J_i is some factor. Now $\langle S_i \rangle$ is magnetization at i^{th} spin site. But due to translation symmetry, $\langle S_i \rangle = m$ & $J_i = J$. Thus the mean field hamiltonian for ising model can be written as,

$$\boxed{\mathcal{H}_{\text{m.F.}} = -H \sum_i S_i} \text{ where, } H = Jm + h$$

$$\Rightarrow m = \tanh (Bh + J\beta m)$$

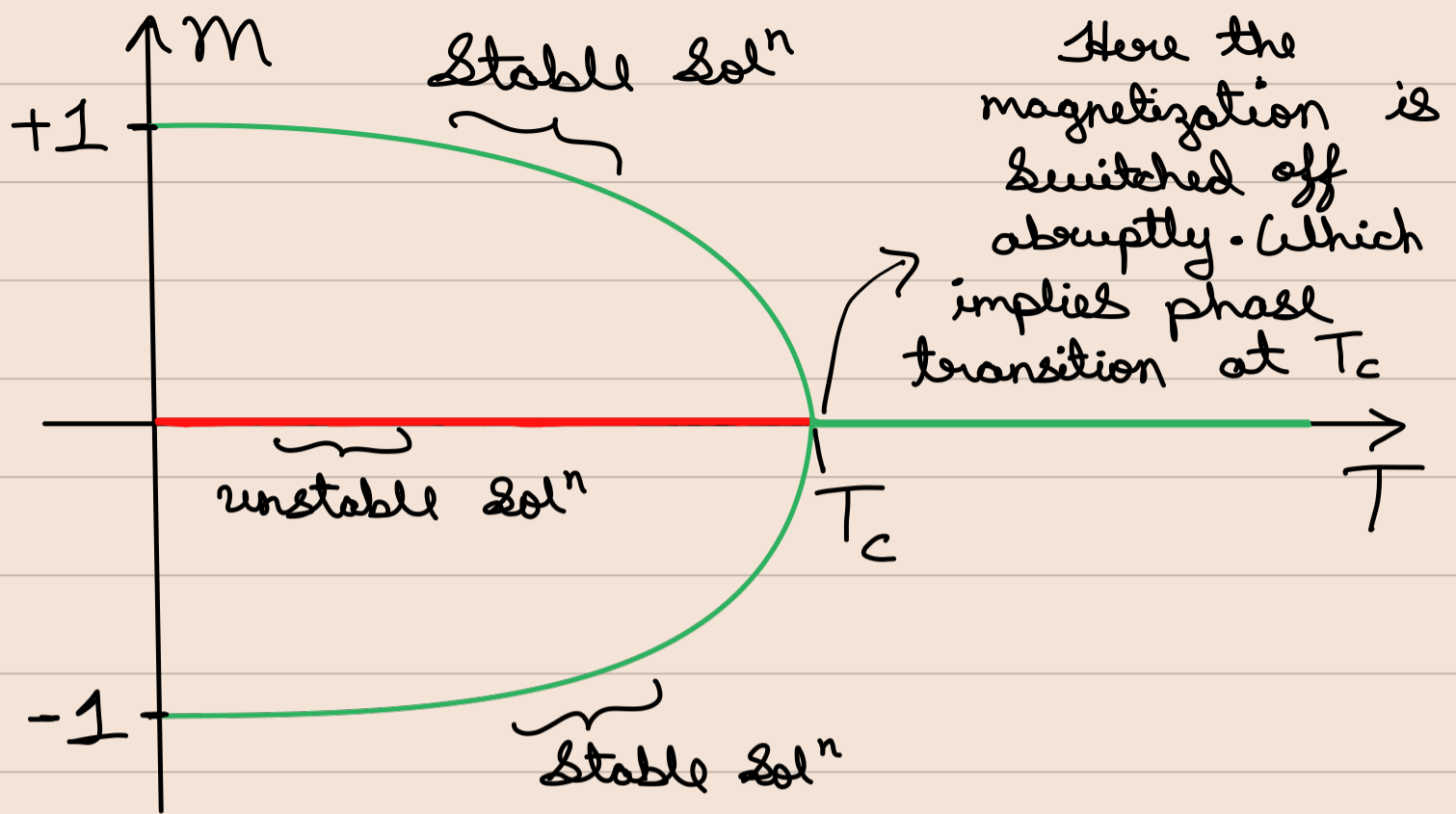
(as solved for ising paramagnet case before)

When $h=0$ (in absence of external field)

$$m = \tanh \left(\frac{Jm}{k_B T} \right)$$

As seen previously we get the critical temperature as $T_c = \frac{J}{k_B}$

$$\Rightarrow m = \tanh \left(\frac{T_c m}{T} \right)$$



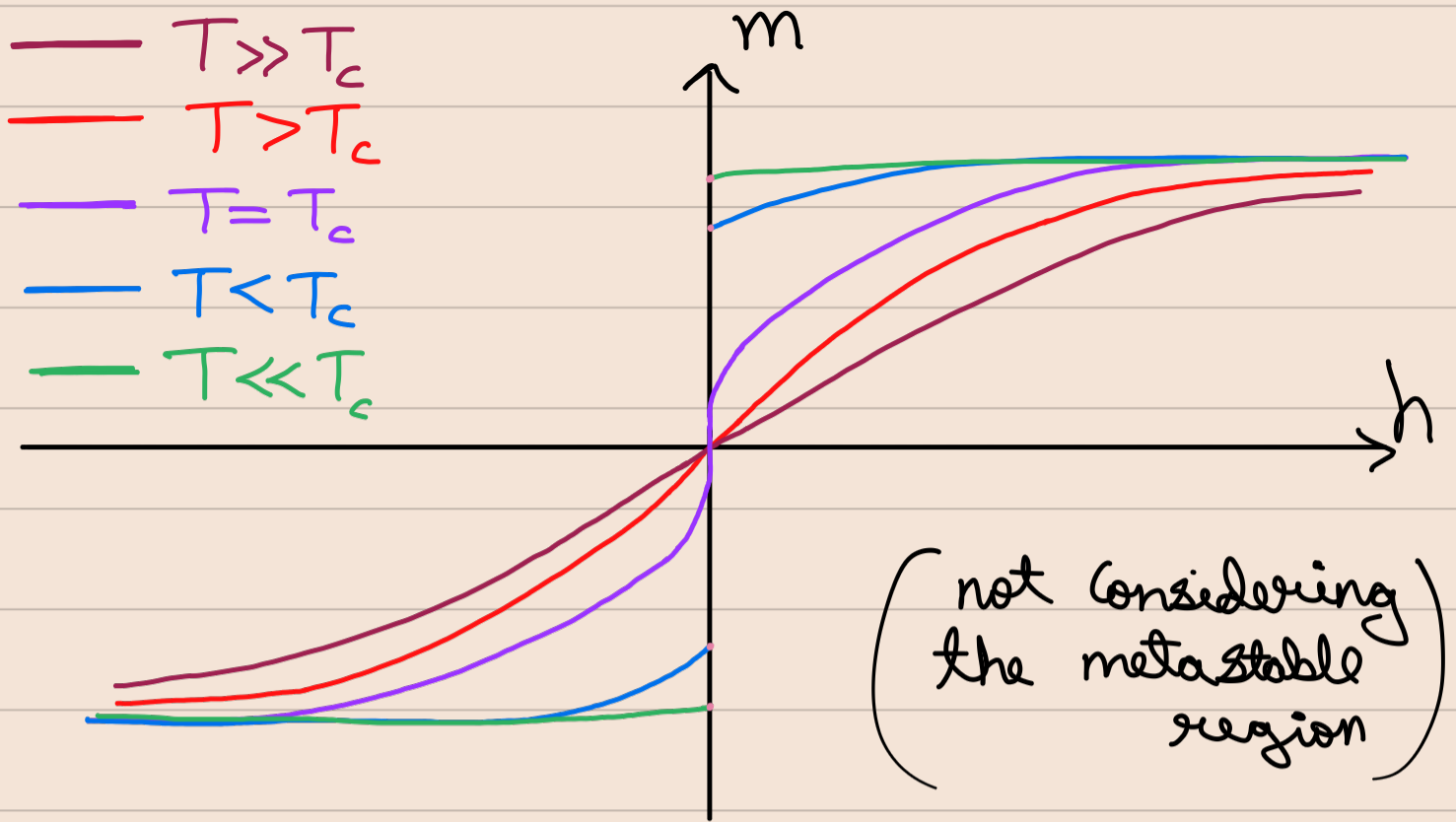
We can also get m vs h curve

Since $m = \tanh(Bh + Bgm)$

$$\Rightarrow \tanh^{-1}(m) = [B(h + gm)] = \frac{h + gm}{K_B T}$$

$$\Rightarrow \tanh^{-1}(m) - \frac{T_c}{T} m = \frac{h}{K_B T}$$

$$\Rightarrow \boxed{h = K_B [T \tanh^{-1}(m) - T_c m]}$$



When $T < T_c$, there will be a sudden flip in the magnetization when the magnetic field reverses its sign.

At $T = T_c$ the slope of $m(h)$ will be infinity w.r.t h .

Since $'m'$ is first derivative of free energy $\Rightarrow$ slope of $'m'$ is the second derivative of free energy.

Since the second derivative is discontinuous / blows out at critical temperature. Thus this reversal of all spins induced by external $'h'$ is a second order phase transition.

$$\text{Now, } \chi_s = \left(\frac{\partial m}{\partial h} \right)_B = \left[\frac{\partial \tanh[B(gm+h)]}{\partial h} \right]_B$$

$$= \text{sech}^2[B(gm+h)] \times \left[g \left(\frac{\partial m}{\partial h} \right)_B + 1 \right] \times B$$

$$\Rightarrow \chi_s = B (g\chi_s + 1) (1 - m^2)$$

$$= Bg(1 - m^2)\chi_s + B(1 - m^2)$$

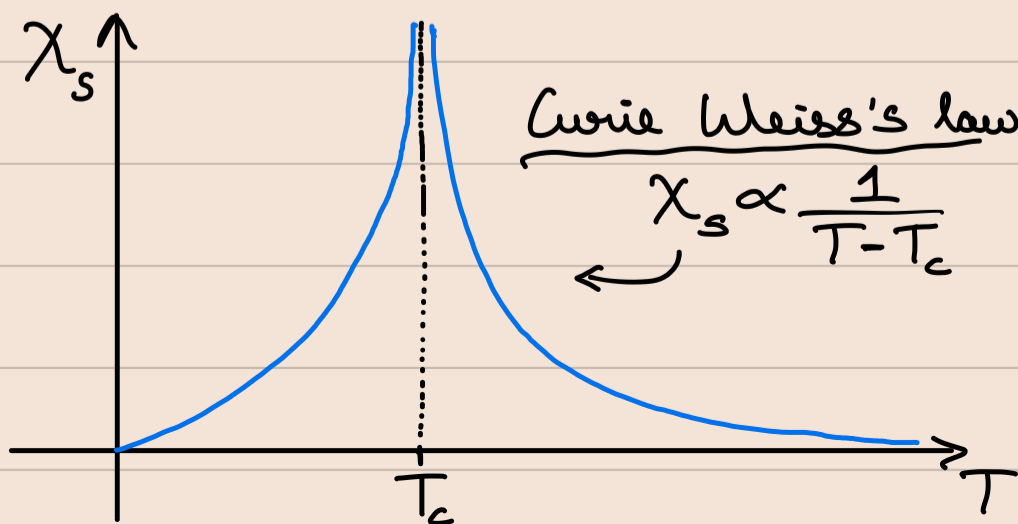
$$\Rightarrow \chi_s \left[1 - \frac{T_c}{T} (1 - m^2) \right] = \left[\frac{(1 - m^2)}{k_B T} \right]$$

$$\Rightarrow \chi_s = \frac{\left[\frac{(1 - m^2)}{k_B T} \right]}{1 - \frac{T_c}{T} (1 - m^2)}$$

$$\text{at } h=0 \Rightarrow m=0$$

$$\Rightarrow \chi_s = \frac{1}{k_B T} \times \frac{1}{1 - \frac{T_c}{T}} = \frac{1}{k_B (T - T_c)}$$

$$\Rightarrow \chi_s \propto \frac{1}{T - T_c} \Rightarrow \text{Curie-Weiss's law}$$



$h=0$

Arithmetic mean - Geometric mean inequality

Let $\{x_i\}$ be a set of random positive values. AM-GM inequality states that,

$$\underbrace{\frac{1}{n} \sum_i x_i}_{A.M} \geq \underbrace{\left(\prod_i x_i \right)^{1/n}}_{G.M}$$

Consider $\{x_1, x_2\}$

$$\Rightarrow (\sqrt{x_1} - \sqrt{x_2})^2 = x_1 + x_2 - 2\sqrt{x_1 x_2}$$

$$\text{Since } (\sqrt{x_1} - \sqrt{x_2})^2 \geq 0$$

$$\Rightarrow x_1 + x_2 - 2\sqrt{x_1 x_2} \geq 0$$

$$\Rightarrow \underbrace{\left(\frac{x_1 + x_2}{2} \right)}_{A.M} \geq \underbrace{\sqrt{x_1 x_2}}_{G.M}$$

Thus $A.M \geq G.M$. This can be proven by mathematical induction for more than two numbers.

Weighted A.M - G.M inequality

Consider $\{p_i\}$ where all p_i are non-negative such that $\sum_i p_i = 1$. For a random non-negative set $X = \{x_i\}$ we can write,

$$\underbrace{\sum_i p_i x_i}_{\text{weighted A.M}} \geq \underbrace{\prod_i x_i^{p_i}}_{\text{weighted G.M}}$$

$$\text{Consider } G = \prod_i x_i^{p_i}$$

$$\Rightarrow \log(G) = \sum_i \log(x_i^{p_i}) = \sum_i p_i \log(x_i)$$

$$\Rightarrow \log(G) = \langle \log(X) \rangle$$

$$\text{Now } A = \sum_i p_i x_i = \langle X \rangle$$

$$\Rightarrow \log(A) = \log(\langle X \rangle)$$

Since $\log$ is a strictly concave function by Jensen's inequality we can say that $\log(\langle X \rangle) \geq \langle \log(X) \rangle$

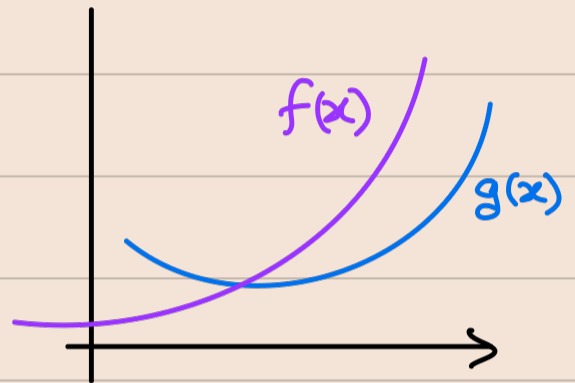
$$\Rightarrow \log(A) \geq \log(G) \Rightarrow A \geq G$$

$$\Rightarrow \boxed{\sum_i p_i x_i \geq \prod_i x_i^{p_i}}$$

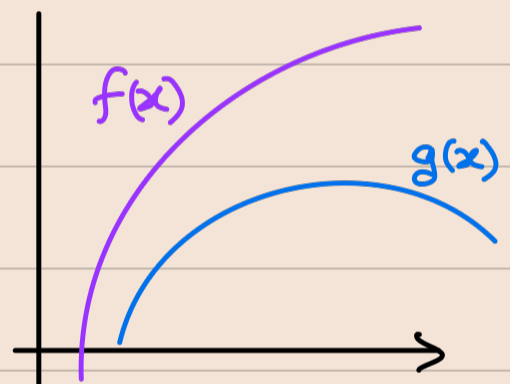
Jensen's Inequality

Jensen's inequality is a general inequality relating expectation values of a function of random variable & the function of expectation value of the random variable.

A function $f(x)$ is convex in the interval (a, b) if $f''(x) > 0$ for all $x \in (a, b)$



A function $f(x)$ is concave in the interval (a, b) if $f''(x) < 0$ for all $x \in (a, b)$



Let φ be a function whose 2nd derivative exists. Let $\langle v \rangle$ denote the expectation value of v . Then by Jensen's inequality,

$$\varphi(\langle x \rangle) \leq \langle \varphi(x) \rangle \Rightarrow \text{if } \varphi \text{ is convex}$$

$$\varphi(\langle x \rangle) \geq \langle \varphi(x) \rangle \Rightarrow \text{if } \varphi \text{ is concave}$$

(equality holds only when either φ is a constant or if x is a constant)

Matrix Inequalities in Stat. Mech

Feynmann's Inequality

Let $\hat{H}$ be the Hamiltonian of the system & $|\psi\rangle$ be any general state (normalized). The expectation value of the Hamiltonian in the state $|\psi\rangle$ is $\langle \hat{H} \rangle = \langle \psi | \hat{H} | \psi \rangle$. And the expectation value of $e^{-\beta \hat{H}}$ is $\langle \psi | e^{-\beta \hat{H}} | \psi \rangle$.

Feynmann inequality is written as

$$\underline{\langle \psi | e^{-\beta \hat{H}} | \psi \rangle} \geq e^{-\beta \langle \psi | \hat{H} | \psi \rangle} \Rightarrow \underline{\langle e^{-\beta \hat{H}} \rangle} \geq e^{-\beta \langle \hat{H} \rangle}$$

Feynmann's inequality is a special case of Jensen's inequality & can be proven using weighted A-M-G-M inequality as follows.

Proof Let $\{|n\rangle\}$ be the eigen states of $\hat{H}$ with eigen value E_n .

$$\Rightarrow \hat{H}|n\rangle = E_n|n\rangle$$

So $\{|n\rangle\}$ is complete. $\Rightarrow \sum_n |n\rangle\langle n| = \hat{I}$

$$\Rightarrow |\psi\rangle = \sum_n (\langle n|\psi\rangle) |n\rangle = \sum_n c_n |n\rangle$$

$$\begin{aligned} \Rightarrow \langle\psi|\hat{H}|\psi\rangle &= \sum_{m,n} \langle\psi|m\rangle \langle m|\hat{H}|n\rangle \langle n|\psi\rangle \\ &= \sum_{m,n} c_m^* \delta_{mn} E_m c_n \end{aligned}$$

$$\Rightarrow \langle\psi|\hat{H}|\psi\rangle = \sum_n |c_n|^2 E_n$$

Let $p_n = |c_n|^2$. It gives the probability that the state $|\psi\rangle$ to collapse to $|n\rangle$ upon measuring. $\Rightarrow \underline{\sum_n p_n = 1}$

$$\Rightarrow \langle\hat{H}\rangle = \sum_n p_n E_n \Rightarrow e^{\langle\hat{H}\rangle} = e^{\sum_n p_n E_n}$$

$$\Rightarrow e^{\langle\hat{H}\rangle} = \prod_n e^{p_n E_n} //$$

Now we need to find expectation value of $e^{-\beta\hat{H}}$.

$$\text{So } \langle\psi|e^{-\beta\hat{H}}|\psi\rangle = \sum_{m,n} c_m^* c_n \langle m|e^{-\beta\hat{H}}|n\rangle$$

Since $\hat{H}|n\rangle = E_n|n\rangle \Rightarrow \underline{f(\hat{H})|n\rangle = f(E_n)|n\rangle}$

$$\text{So } e^{-\beta \hat{H}}|n\rangle = e^{-\beta E_n}|n\rangle$$

$$\begin{aligned}\Rightarrow \langle \psi | e^{-\beta \hat{H}} | \psi \rangle &= \sum_{m,n} c_m^* c_n z_{mn} e^{-\beta E_n} \\ &= \sum_n |c_n|^2 e^{-\beta E_n}\end{aligned}$$

$$\Rightarrow \underline{\langle e^{-\beta \hat{H}} \rangle = \sum_n p_n e^{-\beta E_n}}$$

We can use weighted AM-GM inequality: $\sum_n p_n x_n \geq \prod_n x_n^{p_n} \quad (\sum_n p_n = 1)$

$$\Rightarrow \underbrace{\sum_n p_n (e^{-\beta E_n})}_{\langle e^{-\beta \hat{H}} \rangle} \geq \underbrace{\prod_n (e^{-\beta E_n})^{p_n}}_{e^{-\beta \langle \hat{H} \rangle}}$$

Thus we arrive at Feynmann's inequality which is

$$\boxed{\langle \psi | e^{-\beta \hat{H}} | \psi \rangle \geq e^{-\beta \langle \psi | \hat{H} | \psi \rangle}}$$

Equality holds only if $|\psi\rangle$ is an eigen state of Hamiltonian.

Peierl's Inequality

Let Z be the partition function.
Then Peierl's inequality is,

$$Z \geq \sum_{\nu} e^{-\beta \langle \nu | \hat{H} | \nu \rangle}$$

where $\{|\nu\rangle\}$ are some arbitrary states of the system which need not be complete.

Proof We know that $Z = \text{Trace}[e^{-\beta \hat{H}}]$
 $\Rightarrow Z = \sum_n \langle n | e^{-\beta \hat{H}} | n \rangle$

Here $\{|n\rangle\}$ is complete i.e. $\sum_n |n\rangle \langle n| = \hat{I}$

By Feynmann inequality, we get

$$\langle \nu | e^{-\beta \hat{H}} | \nu \rangle \geq e^{-\beta \langle \nu | \hat{H} | \nu \rangle}$$

$$\Rightarrow \sum_{\nu} \langle \nu | e^{-\beta \hat{H}} | \nu \rangle \geq \sum_{\nu} e^{-\beta \langle \nu | \hat{H} | \nu \rangle}$$

If $\{|\nu\rangle\}$ is not complete, cannot span the entire Hilbert space.

$$\Rightarrow \sum_{\nu} |\nu\rangle\langle\nu| \neq \hat{I}$$

$$\Rightarrow \sum_n \langle n| e^{-\beta \hat{H}} |n\rangle \geq \sum_{\nu} \langle \nu| e^{-\beta \hat{H}} |\nu\rangle$$

$$\Rightarrow Z \geq \sum_{\nu} \langle \nu| e^{-\beta \hat{H}} |\nu\rangle \geq \sum_{\nu} e^{-\beta \langle \nu| \hat{H} | \nu \rangle}$$

$$\Rightarrow Z \geq \sum_{\nu} e^{-\beta \langle \nu| \hat{H} | \nu \rangle}$$

Note: Even if $\{|\nu\rangle\}$ was complete but if $|\nu\rangle$ are not eigen states of $\hat{H}$, then still the Peierls inequality holds as a direct consequence of Feynmann's inequality.

$$Z = \sum_n \langle n| e^{-\beta \hat{H}} |n\rangle = \sum_{\nu} \langle \nu| e^{-\beta \hat{H}} |\nu\rangle$$

where $\{|\nu\rangle\}$ complete basis but not eigen basis of $\hat{H}$.

$$\langle \nu| e^{-\beta \hat{H}} |\nu\rangle \geq e^{-\beta \langle \nu| \hat{H} | \nu \rangle}$$

$$\Rightarrow Z \geq \sum_{\nu} e^{-\beta \langle \nu| \hat{H} | \nu \rangle}$$

Bogoliubov Inequality

Let $\hat{H}$ be the true hamiltonian of the system. And let $\hat{H}_0$ be the trial/approximate Hamiltonian. Then, Bogoliubov inequality is,

$$\underline{F \leq F_0 + \langle \hat{H} - \hat{H}_0 \rangle_0}$$

where $F \Rightarrow$ Actual free energy
 $F_0 \Rightarrow$ Approximate free energy
 $\langle \dots \rangle_0 \Rightarrow$ Average taken w.r.t H_0

$$\text{Eg: } \rightarrow \langle \hat{A} \rangle_0 = \frac{\text{Trace}(\hat{A} e^{-\beta \hat{H}_0})}{\text{Trace}(e^{-\beta \hat{H}_0})}$$

The approximate partition function is

$$\underline{Z_0 = \text{Trace}(e^{-\beta \hat{H}_0})}$$

Usually $\hat{H}_0$ is approximated such that it's eigen states are easier to be found.

Proof Let $\{|\nu\rangle\}$ be the eigenstates of $\hat{H}_0$. By Peierl's inequality,

$$Z \geq \sum_{\nu} e^{-\beta \langle \nu | \hat{H} | \nu \rangle}$$

$$Z \geq \sum_{\nu} e^{-\beta \langle \nu | [(\hat{H} - \hat{H}_0) + \hat{H}_0] | \nu \rangle}$$

$$\Rightarrow Z \geq \sum_{\nu} e^{-\beta \langle \nu | \hat{H} - \hat{H}_0 | \nu \rangle} e^{-\beta \langle \nu | \hat{H}_0 | \nu \rangle}$$

$$\text{Now } \hat{H}_0 |\nu\rangle = E_{\nu}^{(0)} |\nu\rangle$$

$$\& e^{-\beta \langle \nu | \hat{H}_0 | \nu \rangle} = e^{-\beta E_{\nu}^{(0)}}$$

$$\Rightarrow \frac{Z}{Z_0} \geq \sum_{\nu} \left[e^{-\beta \langle \nu | \hat{H} - \hat{H}_0 | \nu \rangle} \right] \frac{e^{-\beta E_{\nu}^{(0)}}}{Z_0}$$

$$\text{Now } \frac{e^{-\beta E_{\nu}^{(0)}}}{Z_0} = p_{\nu} \Rightarrow \text{Prob. of system to be in state } |\nu\rangle.$$

$$\Rightarrow \frac{Z}{Z_0} \geq \sum_{\nu} e^{-\beta \langle \nu | (\hat{H} - \hat{H}_0) | \nu \rangle} p_{\nu}$$

$$\text{Now } \sum_{\nu} e^{-\beta \langle \nu | (\hat{H} - \hat{H}_0) | \nu \rangle} p_{\nu}$$

can be expanded as,

$$\sum_{\nu} \left[1 - \beta \langle \nu | \hat{H} - \hat{H}_0 | \nu \rangle + \frac{\beta^2}{2!} (\langle \nu | \hat{H} - \hat{H}_0 | \nu \rangle)^2 \dots \right] p_{\nu}$$

$$= 1 - \beta \langle \hat{H} - \hat{H}_0 \rangle_0 + \frac{\beta^2}{2!} (\langle \hat{H} - \hat{H}_0 \rangle_0)^2 + \dots$$

$$= \exp \left[-\beta \langle \hat{H} - \hat{H}_0 \rangle_0 \right]$$

$$\Rightarrow \frac{Z}{Z_0} \geq e^{-\beta \langle \hat{H} - \hat{H}_0 \rangle_0}$$

$$\Rightarrow \log(Z) \geq \log(Z_0) - \beta \langle \hat{H} - \hat{H}_0 \rangle_0$$

$$\underbrace{-K_B T \log(Z)}_F \leq \underbrace{-K_B T \log(Z_0)}_{F_0} + \langle \hat{H} - \hat{H}_0 \rangle_0$$

$$\Rightarrow \boxed{F \leq F_0 + \langle \hat{H} - \hat{H}_0 \rangle_0}$$